KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 11 · MATH

The Dairy Farm Schedule

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Factors on the Farm

Nadia lives on a dairy farm in Wisconsin. Before the lesson, warm up with these problems. Show your work.

1 Nadia puts 12 calves into equal rows. Write every multiplication fact that equals 12 (for example, 1 × 12).

2 Skip-count by 4 to write the first six multiples of 4.

3 Which number can be divided evenly by 5?

42 55 63 71

4 List all the factors of 20.

5 Match each word to its meaning. Write the letter next to each word.

___ **factor**	**A.** the answer to a multiplication problem
___ **multiple**	**B.** a number you multiply to get a product
___ **product**	**C.** the answer when you multiply a number by 1, 2, 3, 4…

6 Review (Week 7): 576 ÷ 24 = ?

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 11 · MATH

The Dairy Farm Schedule

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Nadia's Farm Schedule

Use factors, multiples, and prime numbers to finish Nadia's plans. Show your work.

1 Nadia has 42 bales of hay. List every factor pair of 42. Then name one way she could stack the bales in equal stacks.

2 Which cow ID number is prime?

(39) (43) (49) (57)

3 The milk truck comes every 3 days and the feed truck comes every 4 days. Both came today. In how many days will they come on the same day again? What is this number called?

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The Dairy Farm Schedule

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Post-Lesson Challenge: Nadia's Farm Schedule

4. Nadia packs 16 bags of cheese curds and 24 milk bottles into identical crates with nothing left over. What is the greatest number of crates she can fill? How many bags and how many bottles go in each crate?

5. Writing: Nadia's brother says, "87 is prime because it is odd." In two or three complete sentences, explain whether he is right. Use a divisibility rule and the word composite or prime.

★ **Challenge:** The feed truck comes every 4 days, the vet every 6 days, and the cheese buyer every 9 days. All three came on Day 0. On what day will all three come on the same day again?

Looking Ahead: Next week (Week 12) we visit an apartment build in Houston, where factors help us find equivalent fractions, simplify fractions, and work with mixed numbers.

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 11 · READING

The Night the Calf Came Early: Punctuating Dialogue

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Talking on the Page

Before the lesson, show what you already know about writing what characters say. Do your best. It is fine to guess!

1. Add quotation marks around Grandpa's exact words: Grandpa said, The barn is warm.

2. Which sentence is punctuated correctly?

   "Grab your boots," Grandpa said. | "Grab your boots". Grandpa said. | Grab your boots, "Grandpa said."

3. Rewrite with the correct capital letters: nadia said, "the snow is deep."

4. Add the missing comma: "Let's feed the calves" Nadia said.

5. Review: At the start of a story a girl says, "I can't do this." At the end she says, "I did it!" What does this show?

   Character change | Text structure | The author's purpose is to inform

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 11 · READING

The Night the Calf Came Early: Punctuating Dialogue

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Dialogue Editor

Now use every dialogue rule you practiced today. Check quotation marks, commas, capitals, end marks and paragraphs.

1. Rewrite with correct punctuation and capitals: grandpa asked is clover okay

2. Find and fix the TWO mistakes: "we did it", Nadia shouted.

3. This conversation is written as one paragraph. Rewrite it correctly as two paragraphs: "Is the calf warm?" Nadia asked. "Warm as toast," Grandpa said.

4. Nadia is trying not to wake the sleeping calf. Which tag fits best?

   shouted whispered hollered called

5. Writing: Write 4 lines of dialogue between Nadia and Grandpa the next morning when they visit Blizzard. Use a tag BEFORE a quote at least once and AFTER a quote at least once, start a new paragraph for each speaker, and let the dialogue show how one character feels.

★ **Challenge:** Write a short scene (6 to 8 lines) set on a farm, in a city, or in a village where two characters disagree about something. Use dialogue to show each character's viewpoint, start a new paragraph for each speaker, and use only ONE strong tag (like muttered or called) in the whole scene.

Looking Ahead: Week 12: Informative writing. You will build strong paragraphs with topic sentences, supporting details and transition words while touring an apartment building going up in Houston.

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 11 · SCIENCE

Matter and Mixtures

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. What is matter? Name three things that are matter and one thing that is not.

2. Predict: When a log burns down to a small pile of ash, does most of its matter disappear? Where do you think it goes?

3. Draw a mixture you know (a salad, trail mix, a toy box). Label two parts that keep their own properties.

4. Predict: A cup of water is weighed. Then a spoonful of salt is stirred in until it dissolves. Will the cup weigh more, less or the same? Why?

5. What is one thing you wonder about mixing or changing materials?

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 11 · SCIENCE

Matter and Mixtures

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Explain the difference between a mixture and a solution. Give one example of each.

2. Use the class data: why did Trial 3 lose mass but Trial 4 did not? Use the word 'gas' in your answer.

3. Write the steps you would use to separate a mixture of sand, salt and iron filings. Name the property each step uses.

4. Choose Alice Ball or Mario Molina. How did knowing the properties of matter help that scientist help people?

★ **Challenge:** With a parent, put a little baking soda in a bottle and a little vinegar in a balloon. Stretch the balloon over the bottle's top, weigh it, then tip the vinegar in. Weigh it again. Did the mass change? Explain why.

Looking Ahead: Next week we explore energy and electricity: forms of energy, circuits, and why some materials let electricity flow while others block it. Question to bring: Why are electrical wires made of metal on the inside and plastic on the outside?

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 11 · SOCIAL STUDIES

Westward Expansion and Indigenous Removal

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. What do you already know about how the United States grew west after 1800? Write two things.

2. Have you heard of the Trail of Tears, Lewis and Clark, or Sacagawea? Write what you know about one of them.

3. Prediction: If another country said it now owned the land your family lives on, what could your family do to fight back without weapons? Write or draw your ideas.

4. Write one question you have about Indigenous nations in the 1800s.

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 11 · SOCIAL STUDIES

Westward Expansion and Indigenous Removal

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

What Did I Learn?

Use evidence from today's lesson. Write in complete sentences.

1. Why do historians say France sold only a claim to Louisiana in 1803? Who lived on that land?

2. Name two ways the Cherokee Nation fought removal without weapons.

3. Explain why the Cherokee were forced west even after winning Worcester v. Georgia. Use the words "enforce" and "president."

4. Using the Cherokee Nation's 1830 Address as evidence, explain what the Cherokee wanted and what reasons they gave.

★ **Challenge:** Imagine you are writing for the Cherokee Phoenix in 1832, the week after Worcester v. Georgia. Write a short news report that explains the ruling and asks what the president will do next.

Looking Ahead: Next week we travel to Mexico's northern lands, Texas, California and the Gold Rush. Bring this question: what happens to the people who already live in a place when a border moves around them?